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Using two-stream EfficientNet-BiLSTM network for multiclass classification of disturbing YouTube videos

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Abstract
YouTube video recommendation algorithm plays an important role in enhancing user engagement and profitability (or monetization), yet it struggles to deal with disturbing visual content. Real-time analysis using deep learning techniques can play a vital role to identify and filter the disturbing content in online videos. In this paper, we propose an end-to-end trainable two-stream deep learning framework that analyzes and classifies the different categories of disturbing content embedded in child-friendly cartoon videos on YouTube and YouTube Kids platforms. At first, the model extracts the static and motion features from videos through an individual pretrained convolutional neural network (CNN) i.e., EfficientNet-B7. In the next phase, the extracted features are processed by spatio-temporal bidirectional long short-term memory (BiLSTM) network to capture the long-term global temporal dependencies of videos. The learned video representations are forwarded to the classifier for multiclass classification into six categories. Three different types of fusion strategies are investigated to combine the spatial and temporal streams. The evaluation of these methods is performed through extensive experiments on a customized large-scale dataset–YouTube cartoon content filtering (YT-C2F) dataset. The spatio-temporal EfficientNet-BiLSTM network with feature-level fusion displays the best results (f1-score = 0.9316) and shows the efficiency of using a two-stream network compared to single-stream baseline methods. This paper makes two-fold contributions. First, using only static information from videos in a deep learning framework is less effective than using both static and motion information. Second, the feature-level fusion of two streams of networks generates better classification results than early and late fusion techniques. The performance comparison of the proposed model with existing state-of-the-art techniques confirmed the competitive or even better results of our framework.

Keywords Video classification · Deep learning · Convolutional neural network · YouTube

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1 Introduction

Social media is established as an interactive communication media that has been fused into human lives worldwide. Social media platforms including YouTube, Twitter and Facebook have distinctive regulations, architectures, and cultures [65]. As a result, these platforms exhibit different user base interactivity at different times. It depends on the level of sociability and information [7]. YouTube is not just an entertainment platform, but it has evolved into a valuable resource for sharing information and gaining knowledge through videos. It enables direct access to both generic and personalized multimedia content to more than 2 billion monthly active users globally [13, 22]. Younger generations, in particular, are growing up in the era of social media and have adopted YouTube as a viable alternative to traditional education and entertainment tools [53]. The low barrier of restrictions in entry to this platform is the key reason for such high approval ratio among young audiences [54]. The YouTube press release of 2021 confirmed that their platform is highly preferred among youth over other age groups [12].

The possible risks associated with exposure to disturbing content on social media platforms are increasingly serious. Parental control and content filtering (i.e., "restricted mode") mechanisms are applied on both YouTube and YouTube Kids to address the risks of platform reliability and data integrity [9, 16]. Despite these automatic regulations, both platforms face challenges to monitor and regulate user-generated content [43]. These challenges are exploited by malicious uploaders by sharing offensive or objectionable content, particularly in children's videos [48]. The "Elsagate" neologism is referred to those videos on YouTube and YouTube Kids which are depicted as children-friendly or safe but include children-inappropriate scenes or themes [8, 61]. Numerous studies [6, 14, 20, 57, 79] have also reported the Elsagate trend in child-friendly cartoon videos on both YouTube and YouTube Kids platforms. Figure 1 shows some examples of disturbing frames depicted in different cartoon videos from YouTube and YouTube Kids platforms. Livingstone et al. [46] identified that exposing children to obscenity is one of the major internet safety threats for them and the frequent exposure to profanity may have short-term or long-term impact on child's behavior [10]. Hence, the filtering of such sensitive multimedia content should be performed before presenting to certain demographics (i.e., children).

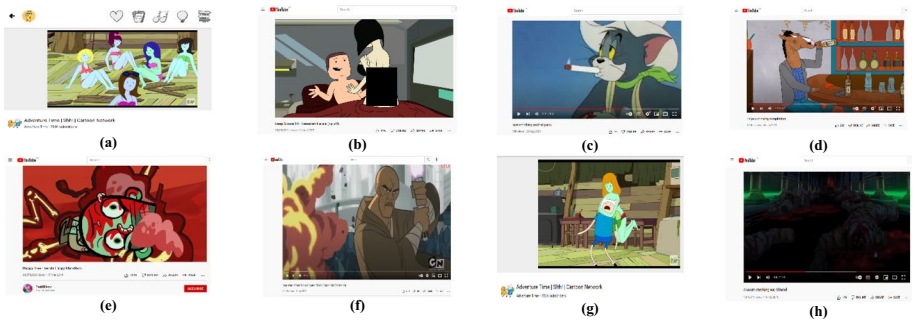


Fig. 1 Examples of cartoon videos on both platforms i.e., YouTube and YouTube Kids. **a**-**b** video clips of the same cartoon on both platforms showing sexual-nudity content. **c**-**d** video clips from different categories of cartoons displaying the act of consuming alcohol, drugs and smoking. **e** video clip containing extreme violence by showing bloodshed. **f**-**g** video clips from both YouTube platforms showing violent acts. **h** the video clip displays an extremely frightening and intense scene

The traditional approach to sensitive content (i.e., pornography, violence) detection applies handcrafted feature extraction techniques such as skin-based [34, 42] and bag-of-visual-words (BoVW) [18, 55] to extract the features of video frames and perform classification. Few studies explored the motion information or spatio-temporal features of videos to disambiguate the complexities in sensitive content classification. In these cases, the evaluated spatio-temporal features are dense trajectories [76], temporal robust features (TRoF) [51, 58], space-time interest points (STIP) [41], and MPEG-motion vectors [33]. In contrast to handcrafted features, deep neural networks have successfully been used in variety of applications to learn the feature representations of videos. Particularly, the convolutional neural network has shown groundbreaking results for image and video classification tasks [39, 52]. CNN-based architectures combining spatial and temporal information are also explored for video classification problems [36, 62]. Moreover, the long short-term memory (LSTM) networks built on top of CNN features are employed to model the long-term temporal clues of videos and achieved competitive performance [81–83].

Regardless of the success of deep neural networks in computer vision, the scarcity of literature is found on the detection and classification of sensitive content from children-oriented videos (such as animated cartoons). When it comes to sensitive visual content filtering, deep learning-based solutions encounter some intriguing challenges, such as: how to design an appropriate deep learning architecture, how to use a pretrained neural network for image/video classification, and what type of visual information is required to understand the visual context. The disturbing behaviors are generally characterized by the body positions, motion and interactions, but appearance-based variables such as color, background, and texture information can also help in sensitive content detection. Considering these factors, a two-stream deep learning framework is proposed to detect and classify disturbing content automatically. The following points are encapsulating the significant contributions of this study:

1. A novel end-to-end trainable two-stream deep learning framework is presented by leveraging a pretrained CNN (i.e., EfficientNet-B7) followed by bidirectional LSTM.
2. A simple video input preprocessing technique is applied in which RGB frames of videos are extracted for the spatial stream and dense optical flows for all points (or pixels) of extracted RGB frames are computed for the temporal stream.
3. Three different fusion strategies are experimented with to combine the static and motion streams of information.
4. The performance of the model is validated on a YT-C2F dataset. The experiment result shows the effectiveness of the proposed model and achieves competitive or even better results than existing state-of-the-art approaches.
5. A ground-truth dataset (YT-C2F dataset) containing 124 hours of cartoon videos from six categories is offered. The videos for the dataset are retrieved from YouTube using YouTube Data API where 100 popular cartoons of different age groups are selected as video search keywords. The dataset used in [85] is the subset of this dataset. For future research, this dataset will also be made available to the scientific community.

The rest of this paper is organized into the following five sections, Section 2 reviews the existing approaches for disturbing content classification problems. Section 3 explains the methodology of the proposed system. The introduction of our tailored dataset, implementation details, ablation study and comparison with existing state-of-the-art approaches are described in Section 4. The hypothesis and limitations of this study are explained in Section 5. The conclusion and some potential directions for future research are described in Section 6.

2 Related work

This section discusses the recent efforts and contributions made by adopting different approaches to disturbing video classification. Several strategies have been presented in the literature to improve the classification results, which are described further below.

2.1 Handcrafted feature extraction techniques

The traditional techniques for sensitive content detection and classification are the construction of handcrafted features to describe the appearance and motion trajectory. The appearance information using skin color has been explored extensively for violence and nudity detection [23–25, 34]. The BoVW frameworks, also referred to sometimes as bag-of-words (BoW) models, accomplished the next milestone in sensitive content detection by combining the keyframe analysis methods with motion analysis methods [18, 47]. Nievas et al. [55] applied STIP and motion scale-invariant feature transform (MoSIFT) feature detection algorithms in the BoW model. Jansohn et al. [33] used skin information with motion histograms in the BoVW framework. Different machine learning classification algorithms including support vector machine (SVM), k-nearest neighbor (kNN) and random forest (RF) are also used to process the appearance and motion trajectories. Jung et al. [35] transformed motion trajectory into a 1-dimensional signal through discrete Fourier transform and handled it with color information in an SVM classifier. Ochoa et al. [56] implemented the sequential minimal optimization (SMO) method during the training of SVM to process the spatio-temporal video features. The study of Tang et al. [72] detected the pornography content from videos by combining the unsupervised clustering in latent Dirichlet allocation (LDA) and supervised learning in SVM models. Lee et al. [42] detected objectionable videos through a multilevel hierarchical system. The three-level system worked with video hash signatures for early detection, single video frame features for real-time detection and group-of-frame features for posterior detection. Kaushal et al. [37] adopted a supervised machine learning technique by feeding the user-level data to the kNN classifier, video-level data to the RF classifier and comment-level data to the decision tree classifier. For violence detection, Hassner et al. [29] extracted the violent flow (ViF) descriptors through optical flow magnitudes and performed binary classification using an SVM classifier. Gao et al. [26] proposed an oriented violent flow (OViF) feature extraction method for violence detection problem. The OViF feature descriptors represent both motion magnitudes and motion orientations, which is further processed through linear SVM and AdaBoost classifiers. Some studies worked with multimodal approach by taking the audio and video features for illicit content detection. Rea et al. [60] extracted visual features through skin segmentation (i.e., color and motion information) and audio stream periodic patterns for illicit video content identification. Liu et al. [44] fused audio vocabulary with visual features for pornography video detection. In this framework, the energy envelope units (EEU) with BoW are applied for audio-vocabulary information extraction. The visual-feature-based method extracts the visual features of keyframes. The fusion method combines audio-visual information, an integrated weighted fusion and a periodicity-based video decision algorithm. Later on, this research is extended by developing the audio vocabulary method using k-means clustering of EEUs [45]. Ulges et al. [74] processed skin color, motion and audio information through three separate SVM classifiers. The weighted sum of late fusion is applied to combine the classification scores of individual classifiers.

2.2 Deep learning techniques

With the advances in deep learning techniques, many neural network frameworks have been proposed for disturbing image/video content classification. The ImageNet pretrained CNN models are applied frequently to detect the disturbing content in videos. Moustafa [52] fused AlexNet and GoogLeNet in an ensemble classifier model to automatically classify pornographic images and videos. Vitorino et al. [75] used GoogLeNet architecture with transfer learning and fine-tuning techniques for sexually exploitative imagery of children (SEIC) content detection. Ishikawa et al. [32] targeted the Elsagate YouTube videos through fine-tuned NASNet model. Aldahoul et al. [3] detected pornographic content from the extracted features of images. This study compared the classification performance of deep learning feature descriptors against handcrafted feature descriptors and observed that feature descriptor using EfficientNet-B7 with an SVM classifier showed better results than other existing techniques. The LSTM network on top of CNN features is applied successfully in several video classification problems [68, 86], and this approach is also explored in disturbing video content detection. Wehrmann et al. [77] performed pornography video classification by training the CNN-LSTM model on the NPDI pornography dataset [11]. Singh et al. [64] obtained the video descriptors through VGG-16 and learnt the video representations from these descriptors by leveraging the LSTM autoencoder framework. Sumon et al. [67] proposed a CNN-LSTM model for violence detection to identify the violent crowd flows in YouTube videos. Ding et al. [19] detected violence in videos using 3D convolutional neural networks.

The two-stream method is also followed in existing studies to involve the static and motion information of videos. Perez et al. [59] performed pornography video detection by employing CNNs for separate static and motion streams. This study showed that the performance of mid-level and late fusion is relatively much better than in early fusion of static and motion streams. Wu et al. [81] captured the short-term patterns through CNNs and the long-term temporal dynamics of videos through LSTMs. Hanson et al. [28] used VGG-13 for the spatial encoder and bidirectional convolutional LSTM (BiConvLSTM) for the temporal encoder to detect violence in videos. Tahir et al. [70] offered the CNN-LSTM model to identify the fake sensitive content integrated into kids' videos on YouTube. Hou et al. [31] worked with multimodal data in a multi-stream network for bloody video recognition. In this network, the spatio-temporal visual features are extracted through the ResNet-LSTM model and audio features are obtained through a CNN model. Ali and Senan [5] employed a simple deep neural network model for violence classification with audio-visual features. Mariconti et al. [49] detected the proactive remarks in YouTube videos by feeding an ensemble classifier with YouTube video metadata, video thumbnail and audio transcripts. Alghowinem [4] and Papadamou et al. [57] proposed deep learning models to detect and filter the Elsagate videos from the YouTube Kids platform.

Based on the findings from related work, it is observed that a lot of research is being done to identify the adult or sensitive content in real-time videos. However, limited to no research has been conducted to analyze the troubling material in children's videos (i.e., animated cartoons). None of the existing studies either performed multiclass classification or included motion information in a deep learning framework to detect the disturbing visual content of cartoon videos which is hiding in plain sight on YouTube and YouTube Kids platforms. Furthermore, it is also noticed that the BiLSTM network is frequently applied in offensive language detection [6, 15, 50, 73, 84], but no study has employed this network for disturbing multiclass video classification problem. The proposed framework extended the

two-stream network by employing EfficientNet-B7 on top of bidirectional LSTM in two separate streams to learn the effective representations from spatio-temporal video descriptors.

3 Proposed method

This study aims to develop a two-stream deep learning framework that is capable of identifying different types of disturbing visual content. The details of the proposed method are described as follows:

3.1 Video preprocessing

The data preprocessing of video is incredibly meaningful as it eliminates noisy data and helps in acquiring the relevant video descriptors. In this paper, the static and motion information from video clips $\{ c_1, c_2, c_3 \ldots, c_{n-2}, c_{n-1}, c_n \}$ are extracted and preprocessed for the two-stream network.

3.1.1 Static information

The spatial stream receives the video static information as input. The static information is often defined by multiple RGB video frames obtained in a fixed or random video-frame sampling interval. Taking the consecutive frames of any video clip requires a lot of computational resources and contains a lot of redundant information. Hence, a constant sampling rate with frame intervals is used to extract the fixed number of RGB video frames with a specific frame distance. The study of Singh et al. [64] also suggested that taking one-fourth of frames from a one-second video clip gives sufficient information by avoiding the frame redundancy problem and computational cost. We selected the frame sampling rate of 6 fps to represent one-fourth information of each video clip (c_i) in the network. Overall, the set of RGB video frames for each video clip (c_i) are represented as $c_i = \{f_1, f_2, \ldots, f_j\}$ where f_j means the jth frame of video clip c_i. The extracted video frame is preprocessed by resizing it into smaller dimensions and maintaining the aspect ratio to fit the default network input dimensions (i.e., 224×224 pixels).

3.1.2 Motion information

Since the motion information of video has demonstrated an enhanced classification performance [36, 62, 74], this study also included motion information by providing the optical flow frames of video clips to the temporal stream of a network. According to RGB video frames in the spatial stream, the corresponding optical flow frames are obtained by estimating the motion of objects between two consecutive RGB video frames. Two types of dense optical flow estimations are evaluated for the proposed network: (1) Gunnar-Farneback [21], and (2) dual TV-L1 [87]. The Gunnar-Farneback technique uses quadratic polynomials to estimate motion between two consecutive RGB frames. It visualizes the direction (or angle) of flow by the value of hue and the magnitude (or distance) by the value of HSV color representation. For optimal visibility, the strength of HSV is set to 255. In dual TV-L1, the motion information is represented by two feature maps i.e., the horizontal component of motion (dx) and the vertical component of motion (dy). These motion maps are transformed into two grayscale images

(i.e., one for each component of motion) through image normalization to [0, 255] interval. Figure 2 shows the generated dual TV-L1 and Gunnar-Farneback dense optical flow estimations of an example video clip.

As mentioned in Section 3.1.1, the spatial stream processes six RGB video frames with dimensions of 224×224 for each video clip. It produces a stack of six grayscale optical flow frames (i.e., three x-channel images and three y-channel images) with dual TV-L1, and denoted as $c_i(dual_tv_l1) = \left\{ f'_{x_1}, f'_{y_1}, f'_{x_2}, f'_{y_2} \ldots, f'_{x_j}, f'_{y_j} \right\}$. The Gunnar-Farneback motion estimation algorithm, on the other hand, produces three colored (RGB) optical flow frames. It is denoted as $c_i(gunnar_farneback) = \left\{ \hat{f}_1, \hat{f}_2, \ldots, \hat{f}_j \right\}$. The output dimensions of optical flow frames are represented as 224×224 x channels (channel = 1 in TV-L1, or channel = 3 in Gunnar-Farneback).

3.2 Spatio-temporal descriptors

To extract the spatio-temporal features, this study followed the approach of [62] where video clips are decoupled into spatial and temporal streams modeled by two individual CNNs. Instead of applying a CNN model from scratch, two pretrained CNN models with a

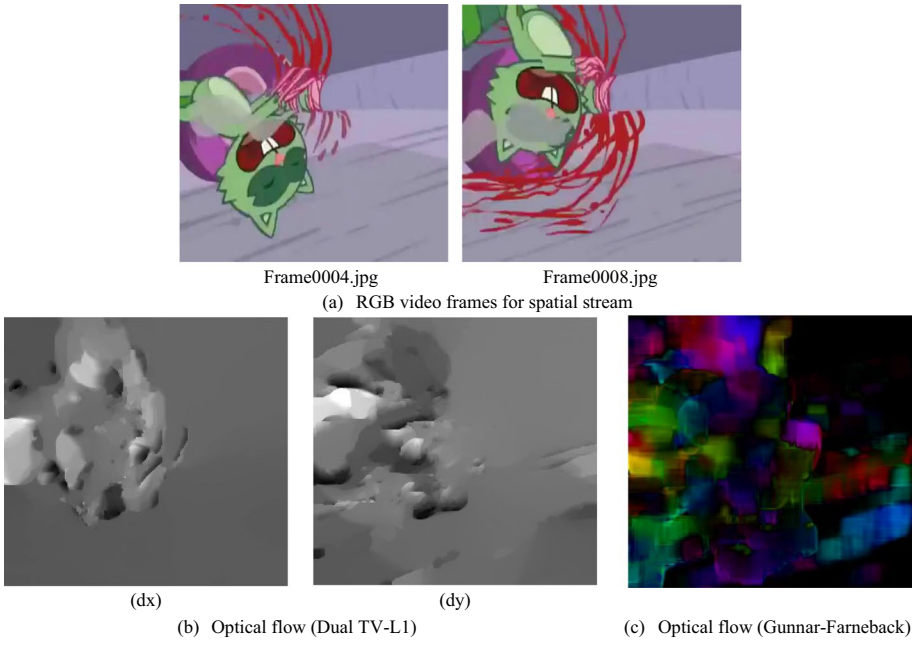


Fig. 2 Samples of RGB and optical flow video representations. The example video clip has a frame rate of 24 fps. Considering our frame sampling rate (i.e., 6fps) with fixed frame interval, every 4th video frame is taken for the spatial stream of network. The optical flow is estimated from every two consecutive RGB video frames used in the spatial stream. **a** RGB video frames are extracted from the video clip (c_i). **b** the grayscale horizontal (dx) and vertical (dy) optical flow frames are obtained by applying dual TV-L1 on RGB frames illustrated in (a). **c** the colored optical flow frame is obtained by applying Gunnar-Farneback on RGB frames illustrated in (a)

transfer learning approach are employed. The EfficientNet model demonstrated better performance and efficiency than existing pretrained CNNs, hence the highest scaled version of baseline EfficientNet known as EfficientNet-B7 is used to individual spatial and temporal streams to extract the spatio-temporal features.

3.2.1 EfficientNet

EfficientNet is a family of convolutional neural networks with a compound scaling method that leverages compound coefficients to uniformly scale the network depth, width and resolution in a simple but effective manner. The architecture of the baseline EfficientNet (EfficientNet-B0) uses mobile inverted bottleneck convolution (MBConv) blocks. These MBConv blocks are as same as in MobileNetV2, except the EfficientNet-B0 has slightly bigger blocks due to larger floating-point operations per second (FLOPS) target. The EfficientNet-B0 model is scaled up to obtain the family of EfficientNets (i.e., EfficientNet-B1 to EfficientNet-B7). Inspired by the performance of EfficientNet-B7, as reported by Tan and Le [71] on ImageNet [17] and CIFAR-100, we employed two EfficientNet-B7 as feature descriptors in spatial and motion streams. As shown in Fig. 3, the EfficientNet-B7 contains several blocks of MBConv layers with different kernel sizes. The input to the model is RGB video frames for the spatial stream EfficientNet and stacked optical flow frames for the motion stream EfficientNet. At the end of the two-stream pipeline, the spatial $(x_1^s, x_2^s, \ldots, x_T^s)$ and motion $(x_1^m, x_2^m, \ldots, x_T^m)$ feature sets are obtained.

3.3 Temporal modeling with BiLSTM

Recurrent neural network (RNN) model can remember the long-term temporal dynamics of input sequences. However, vanilla RNN suffers from vanishing or exploding gradient problems during backpropagation through time. This limitation is resolved through a variant of RNN i.e., long short-term memory network [30]. By default, the conventional LSTM processes input in a forward direction by ignoring the future context. To obtain a long range of information from past and future contexts, Graves and Schmidhuber [27] proposed a BiLSTM network for capturing the contextual dependencies in both directions at the same time with forward and backward hidden layers. A BiLSTM consists of two LSTM units with their hidden layers connected in opposite directions. Each LSTM block at time t receives the output h_{t-1} and cell information c_{t-1} of previous time step $t-1$ through recurrent connections. This block at time step t gets an input vector x_t, and goes separately through each gate together with h_{t-1}. The complete structure of single LSTM cell is illustrated in Fig. 4.

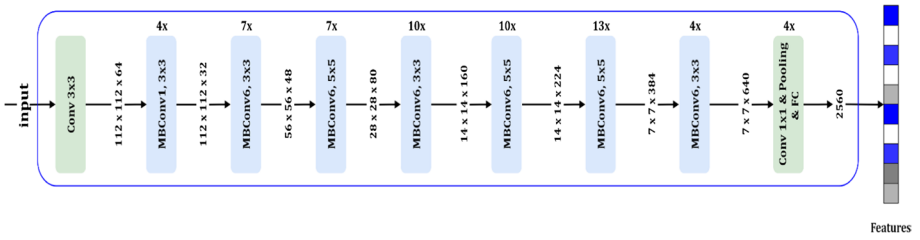


Fig. 3 The network architecture of the EfficientNet-B7 model

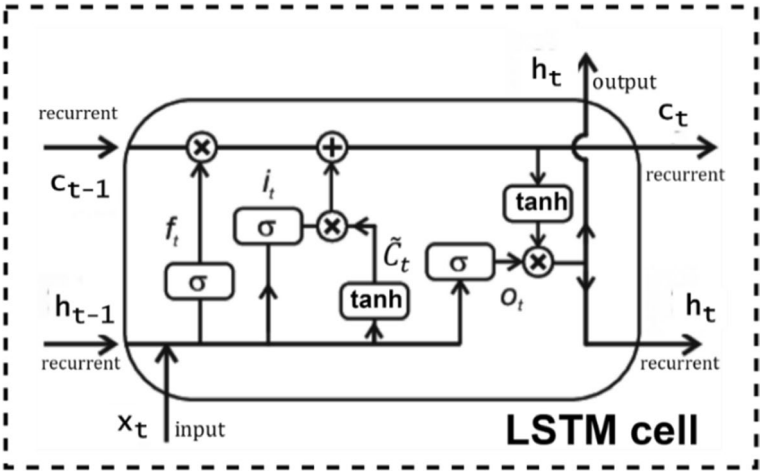


Fig. 4 The structure of LSTM cell

The outputs from forget gate f_t, input gate i_t, memory cell state $\widetilde{C}_t$ and output gate o_t at time t are computed according to Eqs. (1) to (4). The LSTM block output h_t, as mentioned in Eq. 5, is figured out by activating the cell $\widetilde{C}_t$ with *tanh*() activation, and in this the output gate o_t decides how much to give out as an output of the LSTM block.

$$i_t = \sigma\left(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}\widetilde{C}_{t-1} + b_i\right) \tag{1}$$

$$f_t = \sigma\left(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}\widetilde{C}_{t-1} + b_f\right) \tag{2}$$

$$\widetilde{C}_t = f_t\widetilde{C}_{t-1} + i_t\tanh\left(W_{xc}x_t + W_{hc}h_{t-1} + b_c\right) \tag{3}$$

$$o_t = \sigma\left(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}\widetilde{C}_t + b_o\right) \tag{4}$$

$$h_t = o_t.\tanh\left(\widetilde{C}_t\right) \tag{5}$$

In BiLSTM, as shown in Fig. 5, the outputs of the forward layer $\left(\overrightarrow{h_t}\right)$ and hidden sequences are iteratively computed using input vectors x_t in ascending order i.e., ., $t=1$, 2, …, T, while the outputs of the backward layer $\left(\overleftarrow{h_t}\right)$ and hidden sequences are iterated in descending order i.e., $t=T$, $T-1$, …, 1. The outputs from forward and backward hidden layers are computed according to Eqs. (6) and (7):

$$\overrightarrow{h_t} = \overrightarrow{LSTM}\left(x_t\right) \tag{6}$$

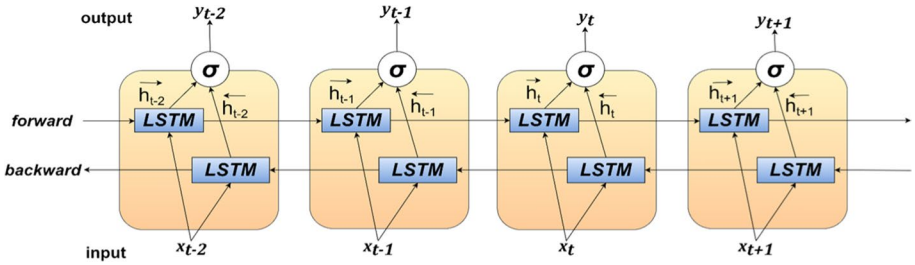


Fig. 5 Bidirectional LSTM model

$$\overleftarrow{h_t} = \overleftarrow{LSTM}(x_t) \tag{7}$$

The BiLSTM layer yields an output vector y_t, which is computed using Eq. (8).

$$y_t = \sigma\left(\overrightarrow{h_t}, \overleftarrow{h_t}\right) \tag{8}$$

where, σ function combines the outputs of forward and backward hidden layers. This σ function can be a concatenation, average, multiplication or summation function. In the proposed two-stream pipeline, the learning process obtains the bidirectional long-term global temporal dependencies through vector sequences of spatial $\left(x_1^s, x_2^s, \ldots, x_T^s\right)$ and temporal $\left(x_1^m, x_2^m, \ldots, x_T^m\right)$ feature sets.

3.4 Fusion strategies

This study has experimented with three different fusion strategies to combine the static and motion information of videos. These fusion strategies of spatial and temporal streams make three different variants of the proposed model. The details of fusion strategies are described below:

3.4.1 Early fusion

In the early fusion method, the static (RGB video frames) and motion (optical flow frames) information are integrated before proceeding through the CNN pipeline for feature extraction. For each video clip, the RGB video frames with respective motion representations (i.e., *dx* and *dy* for dual TV-L1 or colored optical flow frame for Gunnar-Farneback) are stacked together for passing as an input to the EfficientNet-B7 model for feature extraction.

3.4.2 Feature-level fusion

The feature-level fusion is also known as intermediate or mid-level fusion. In this, spatio-temporal feature representations from EfficientNet-B7 followed by BiLSTM network are concatenated before feeding to the classifier block.

3.4.3 Late fusion

In late fusion, the classifiers are processed through separate spatial and motion streams where the classification scores are combined through an average or weighted average method to obtain the final video classification score.

3.5 Classifier

The classifier block consists of two fully-connected layers. The first fully-connected layer with 1024 hidden units utilized rectified linear unit (ReLU) as an activation function, while the second fully-connected layer with 6 output units applied the softmax activation function. The dropout layer is added between these two fully-connected layers in which 30% of neurons are dropped while training.

3.6 Network architecture

The network architecture of the proposed framework consists of spatial and temporal streams, as illustrated in Fig. 6.

In the architecture, video clips are preprocessed to extract static and motion information. The static information (i.e., RGB video frames) is used as an input to the spatial stream, while the motion information (i.e., stacked optical flow frames) is used as an input to the temporal stream. For feature extraction, each stream contains a time-distributed EfficientNet-B7 to handle the RGB and their respective optical flow video frames. In EfficientNet-B7, the last three layers including the output layer are replaced with the global average pooling 2D layer. The global average pooling technique reduces the dimensionality of vectors for the BiLSTM network without causing any loss in network accuracy. The extracted features are passed into two layers of BiLSTMs to capture the bidirectional long-term global temporal dependencies of video clips. Each BiLSTM layer of the spatial stream contains 512 nodes (i.e., 256 LSTM units per direction), while the temporal stream BiLSTMs have 128 nodes (i.e., 64 LSTM units per direction). The learned spatio-temporal representations are fused in concatenate layer and forwarded to the classifier block of two fully-connected layers. The first fully-connected layer of 1024 units with ReLU activation is simply used to learn the non-linear combinations of features, whereas the second fully-connected layer is considered as an output layer of 6 units with a softmax activation function. The dropout layer with a dropout rate of 0.3 is added in between two fully-connected layers to avoid overfitting problems. Finally, the categorical cross-entropy loss function calculates the outputs of the last layer. The complete steps for the training process of the proposed framework are described in Algorithm 1.

4 Experiments and results analysis

4.1 Dataset

The growth of multimedia content on YouTube has prompted researchers to investigate it for a variety of applications. Numerous benchmark datasets are created and exploited using different YouTube data modalities [36, 38, 40, 66, 80]. Google published YouTube-8M [2],

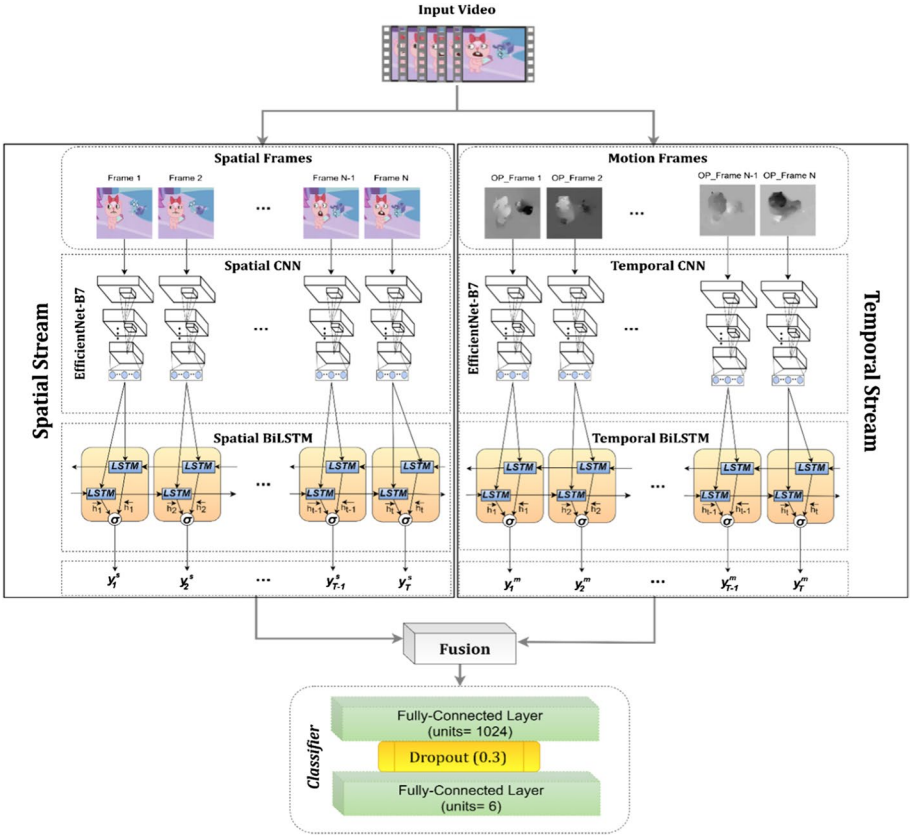


Fig. 6 An overview of the proposed two-stream deep learning framework for disturbing video content classification. Given an input video clip, two types of features are extracted using EfficientNet-B7 (pretrained CNN) models from spatial RGB video frames and stacked optical flow motion frames, respectively. These spatial and temporal features are fed separately into two layers of BiLSTM models for bidirectional long-term global temporal dependencies. The outputs of two feature representations are fused via concatenate layer. Lastly, the classifier block with two fully-connected layers is added to generate the final classification probabilities of each class

a dataset with over 8 million YouTube annotated videos. This extensive dataset features pre-filtered videos from a wide range of categories (i.e., 4716 classes) by excluding adult and sensitive visual content. So, there is no significance in using it for the proposed video classification problem. As a result, other options were explored, including (1) finding the publicly available cartoon-specific dataset, or (2) creating a custom dataset. The closest available datasets used in previous studies on cartoons are the Elsagate dataset [32], the multiclass YouTube cartoon video dataset [64] and the NPDI cartoon dataset [3]. These datasets are either smaller in size or accompanied by less complex behaviors which may not be helpful in the model training process. Therefore, a tailored dataset named YT-C2F is created. The data is collected by preparing a list of 100 cartoon titles as keywords. These keywords retrieve the video IDs from YouTube via YouTube Data API. The US parental guidelines [78] are followed where the cartoon titles from six TV ratings (i.e., TV-Y, TV-G, TV-Y7, TV-Y7-FV, TV-PG, and TV-14) are selected. Based on the popularity and

Input:

Training set $T = \{(c_i, y_i),\ i = 1,2,\ldots,n\}$, Validation set $V = \{(c_j, y_j),\ j = 1,2,\ldots,\hat{n}\}$ where c_i = input training video clip, y_i = assigned label for training video clip (c_i), c_j = input validation video clip, and y_j = assigned label for validation video clip (c_j); Mini-batch of training set $t = \{t_k, \forall\ t_k \in T \mid t_1 \cap t_2 \ldots \cap t_m = \emptyset, k = 1,2,\ldots,m\}$, Mini-batch of validation set $v = \{v_k, \forall\ v_k \in V \mid v_1 \cap v_2 \ldots \cap v_{\hat{m}} = \emptyset, k = 1,2,\ldots,\hat{m}\}$; $total_epochs$ = total number of epochs

Output: Trained model $Model_{epoch}$, Confusion matrix $conf_matrix_{epoch}$

for $epoch = 0, total_epochs$ **do**
  for all $t_k \in T\ (k = 1,2,\ldots,m)$ **do**
    Initialize the $feature_set_{rgb}$ $\boldsymbol{f_{t_k}}$ *and* $feature_set_{optical}$ $\hat{\boldsymbol{f}}_{\boldsymbol{t_k}}$
    for all $c_i \in t_k \subseteq T\ (i = 1,2,\ldots,n_1)$ **do**
      Extract $frames_{rgb}(f_1, f_2, \ldots, f_n)$ *and* $frames_{optical}\ (\acute{f}_1, \acute{f}_2, \ldots, \acute{f}_n)$ *of video clip* (c_i)
      Extract spatial features $(x_1^s, x_2^s, \ldots, x_n^s)$ *of* $frames_{rgb}\ (f_1, f_2, \ldots, f_n)$ *using* $EfficientNet - B7$
      Append $feature_set_{rgb}$ f_{t_k} *with spatial features* $(x_1^s, x_2^s, \ldots, x_n^s)$ *of video clip* (c_i)
      Extract motion features $(x_1^m, x_2^m, \ldots, x_n^m)$ *of* $frames_{optical}\ (\acute{f}_1, \acute{f}_2, \ldots, \acute{f}_n)$ *using* $EfficientNet - B7$
      Append $feature_set_{optical}$ $\hat{f}_{t_k}$ *with motion features* $(x_1^m, x_2^m, \ldots, x_n^m)$ *of video clip* (c_i)
    end for
    Train spatio $-$ *temporal* $EfficientNet - BiLSTM$ *model for batch* t_k : $\boldsymbol{Model_{epoch}.fit((f_{t_k}, \hat{f}_{t_k}), y_{t_k})}$
  end for
  for all $v_k \in V\ (k = 1,2,\ldots,\hat{m})$ **do**
    Extract $frames_{rgb}(f_1, f_2, \ldots, f_n)$ *and* $frames_{optical}\ (\acute{f}_1, \acute{f}_2, \ldots, \acute{f}_n)$ *of validation set video clip* (c_j)
    Repeat steps $6 - 7$ *to extract spatial features and append in validation* $feature_set_{rgb}$ $\boldsymbol{f_{v_k}}$
    Repeat steps $8 - 9$ *to extract motion features of video clip* (c_j) *and append in validation feature set* $\hat{\boldsymbol{f}}_{\boldsymbol{v_k}}$
    Validate the model for epoch : $\boldsymbol{result_{epoch} \leftarrow Model_{epoch}.evaluate((f_{v_k}, \hat{f}_{v_k}), y_{v_k})}$
  end for
  Generate confusion matrix: $conf_matrix_{epoch} \leftarrow Confusion_Matrix(result_{epoch}.predicted, result_{epoch}.actual)$
  Calculate $precision_{epoch}, recall_{epoch}, accuracy_{epoch}$ *and* $f1\text{-}score_{epoch}$ *using* $conf_matrix_{epoch}$
end for
return $Model_{epoch}$, $conf_matrix_{epoch}$

Algorithm 1 Training process of the proposed model

rating scores, the top 25 cartoons in each category are shortlisted. The purpose of selecting different categories is to capture a wide range of video content and genres. In the next step, the first 100 video IDs against each keyword are obtained through the search query of YouTube Data API. This approach resulted in 8790 unique videos with different time durations. The annotation of videos is performed in three rounds to get the labeled data. In the first round, all videos containing information other than cartoons (i.e., cartoon reviews by people, abridged cartoon videos or non-cartoon videos, etc.) are discarded. Next, the FFmpeg split video function divided videos into one-second duration clips. The main reason for splitting videos is to implement multiclass video classification and assigning one class to a full-duration video would not represent the accurate context. In the last round, each one-second video clip is manually reviewed and labeled against one of the six defined classes—safe (S), alcohol, drugs and smoking (ADS), extreme violence (EV), fantasy violence (FV), frightening and intense scenes (FIS), and sexual and nudity (SN). All these classes are defined by adopting the US parental TV guidelines for cartoon videos. Table 1 explains the criteria used to assign a video clip to a specific class. The video clips containing information other than the defined classes are not included in the dataset.

The complete annotation process is carried out by more than 100 undergraduate students (aged 20–22, both male and female) under the supervision of the authors. The entire distribution of the YT-C2F dataset in six classes is shown in Table 2. Overall, the filtering and annotation process resulted in 446,793 video clips (approximately 124 hours of video data). The dataset displays the balanced distribution of safe and unsafe video content (1: 1.12). This dataset will also be made available to the scientific community to further continue the research and generate new insights based on recent deep learning advances.

4.2 Training strategy

As it is noticed that the consecutive frames of video clips contain duplicate information, hence one-fourth information is extracted. The constant sampling rate for RGB video frames is 6 fps by considering the average frame rate (i.e., 23–24 fps) of video clips in the YT-C2F dataset. The random-sized RGB video frames are preprocessed to convert into fixed-shaped frames with the dimensions of $224 \times 224 \times 3$. The dual TV-L1 and Gunnar-Farneback dense optical flow methods are used to calculate the motion information of corresponding RGB video frames. Overall, the video preprocessing stage produces the RGB video frames of shape $6 \times 224 \times 224 \times 3$ and optical flow frames of shape $6 \times 224 \times 224 \times 1$ (in dual TV-L1) or $3 \times 224 \times 224 \times 3$ (in Gunnar-Farneback). Due to the limited memory and computational resources, the batch of 900 video clips from the dataset is loaded simultaneously to fit in memory. In this study, the complete training dataset is processed in 348 iterations per epoch. Furthermore, the variation of gradient descent known as the mini-batch gradient descent optimization algorithm is used in the model training process which further divides the memory-loaded batch of training dataset into n number of mini-batches. The mini-batch size is restricted to 8 due to memory constraints. To summarize, it takes 113 steps to complete one iteration where the model coefficients are updated. The proposed two-stream deep learning framework is implemented using the TensorFlow Python library [1] with Jupyter notebook. All deep learning models are trained on Intel Core i7–7700 @ 2.80GHz with Nvidia GeForce GTX 1050 6 GB graphic card and 32 GB RAM. The training of all models is performed for 50 epochs or until the training model starts overfitting. Furthermore, the dataset is distributed into 70% for training data, 20% for validation data

Table 1 Criteria for video-class labeling

Class	Criteria
Safe	Video content is appropriate or safe for all age groups of children.
Alcohol, Drugs and Smoking	Video content in which cartoon characters are consuming alcohol, smoking cigarettes, taking or sniffing drugs or other related substances.
Extreme Violence	Video content displaying extreme graphic violence or distress such as bloodshed, suicide, torture, assault, intense, gory actions or situations.
Fantasy Violence	Video content displaying violent situations such as characters fighting, beating or hitting one another, using weapons or explosives or engaging in fantasy-based aggressive events.
Frightening and Intense Scenes	Video content portrayal of horror, suspense, or danger through bizarre characters or situations that may frighten younger audiences.
Sexual and Nudity	Video content displaying graphic nudity or sexually illicit actions or situations.

Table 2 Summary of the YT-C2F dataset

Category	Class	Class Keyword	Label	# Clips	Dataset Distribution Ratio (video clips)	Train/Validation/Test dataset size
Cartoons	Safe	S	0	210,313	1:1.12 (Safe: Unsafe)	70:20:10 (train: validate: test)
	Alcohol, Drugs and Smoking	ADS	1	31,269		
	Extreme Violence	EV	2	51,403		
	Fantasy Violence	FV	3	51,269		
	Frightening and Intense Scenes	FIS	4	51,285		
	Sexual and Nudity	SN	5	51,254		
Total video clips					446,793 (approx. 124 hours)	

and 10% for testing data. With our machine configurations, it takes an average of 240 seconds to complete one iteration of the proposed model and an estimated 23.2 hours are required to complete one epoch of training data. The hyperparameters are tuned for the selection of the number of BiLSTM layers and the number of LSTM units (per direction) in each layer. The training of the proposed framework is started with a learning rate of 1×10^{-3}. After every 10 epochs, the learning rate is reduced until it reaches the rate of 1×10^{-5}. Other hyperparameters such as dropout rate, optimizer, and batch size are selected through experiments. Table 3 enlists the values of all the hyperparameters used in the proposed model.

4.3 Results and discussion

4.3.1 Single-stream network with CNN architecture

Different CNN architectures with individual stream settings are evaluated to report the performance of these networks in the devised framework (see spatial and temporal streams of Fig. 6, in this case, both streams are evaluated separately till the CNN modules). The pretrained CNN models for image and video classification are employed to extract features from videos. For RGB video frames, the Inception-V3 [69], VGG-19 [63] and EfficientNet-B7 [71] models are selected. Each model has different number of layers (depth) and trainable parameters. The concept of transfer learning-based CNN is implemented by substituting the last three layers of architectures with an output layer of six outputs using the softmax activation function. According to the results in Table 4, the Inception-V3 model shows relatively lower accuracy than VGG-19 and EfficientNet-B7 models. It is worth noticing that depth-oriented neural networks work better and converge faster on the YT-C2F dataset. Hence, EfficientNet-B7 is selected for the RGB video frames. The dual TV-L1

Table 3 The hyperparameters in the proposed two-stream deep learning model

Parameters	Value
Video file format	MPEG-4
Input RGB frames	6
Input RGB frame dimensions	$224\times224\times3$
Input optical flow frames	6 (Dual TV-L1), 3 (Gunnar-Farneback)
Input optical flow frame dimensions	$224\times224\times1$ (Dual TV-L1) $224\times224\times3$ (Gunnar-Farneback)
No. of BiLSTM layers (static stream)	2
No. of units in BiLSTM layers (static stream)	256 LSTMs per direction
No. of BiLSTM layers (motion stream)	2
No. of units in BiLSTM layers (motion stream)	64 LSTMs per direction
Dropout rate	0.3
Optimizer	Adam
Learning rate	1×10^{-5}
Loss function	Categorical cross-entropy
Mini-batch gradient descent optimization size	8
Training epochs	50

Table 4 Performance of single-stream CNN models on the YT-C2F dataset

Model	Information	Total parameters	Depth	Accuracy (%)
Inception-V3	Static (RGB video frames)	23 Million	159	90.30
VGG-19		143 Million	26	91.47
EfficientNet-B7		66 Million	816	91.59
EfficientNet-B7	Motion (Optical flow frames)	66 Million	816	88.75
DenseNet-201		20 Million	201	84.52

method is applied to get the horizontal and vertical motion maps in the form of two gray-scale video frames for optical flow estimation. Besides EfficientNet-B7, the DenseNet-201 model is experimented with by following the study of [88] to extract the motion features from optical flow frames. The fact that EfficientNet-B7 displayed superior performance than DenseNet-201 on optical flow video frames supported the argument for using it in individual spatial and temporal streams of the neural network framework.

4.3.2 Single-stream vs. multi-stream network with CNN architecture

The performance of the two-stream EfficientNet-B7 network is evaluated and compared with the individual stream settings (i.e., using either static or motion stream of network), and the results are mentioned in Table 5. Apart from dual TV-L1, the Gunnar-Farneback dense optical flow method is implemented to calculate the successive optical flow of consecutive RGB video frames. According to experimental results, the dual TV-L1 method shows better accuracy performance in both individual and two-stream networks, but this is a computationally expensive solution. The dual TV-L1 method takes twice or more time than the Gunnar-Farneback to update the model coefficients in each iteration. Moreover, it is observed that the EfficientNet with transfer learning works relatively better in a static stream by displaying higher accuracy results than the two-stream network configurations. Hence further investigation is carried out by evaluating some other neural networks on top of EfficientNet feature extractor in spatial and temporal (or motion) streams of the deep learning framework.

Table 5 Performance of single-stream vs. two-stream CNN models on the YT-C2F dataset

Model	Approach		Learning rate	Training time per iteration (sec)	Accuracy (%)
EfficientNet-B7	Static	RGB	1×10^{-5}	165	91.59
	Motion	Dual TV-L1	1×10^{-5}	300	88.75
		Gunnar-Farneback	1×10^{-5}	140	85.50
	Static + Motion (Fusion)	RGB + Dual TV-L1	1×10^{-5}	720	89.45
		RGB + Gunnar-Farneback	1×10^{-5}	257	87.73

4.3.3 Multi-stream network with CNN-LSTM and CNN-BiLSTM

The traditional long short-term memory networks on top of convolutional features have already been proven as a successful combination in multi-stream video classification architectures [81–83]. This study also worked with LSTM to understand the long-term temporal dependencies of videos. Two sets of experiments are performed, one with LSTM and the other with BiLSTM on top of CNN (i.e., EfficientNet-B7). The spatial and temporal streams of these deep learning frameworks are combined using different fusion strategies as explained in Section 3.4. The network architecture of EfficientNet-LSTM is identical to the illustrated EfficientNet-BiLSTM framework shown in Fig. 6, with the only difference being the replacement of BiLSTM with LSTM. This way the spatial stream contains two LSTM network layers with 256 hidden units in each layer, while the temporal stream contains two LSTM network layers with 64 hidden units in each layer. Due to limited computational resources, the Gunnar-Farneback method is used to estimate the motion information of videos in the form of optical flow frames. These frames are processed through temporal streams of a network. The evaluation results, as mentioned in Table 6, show that overall two-stream network performance improves when using BiLSTM with different fusion strategies as these models capture the long-term temporal dependencies of videos by understanding the past and future context of videos.

It is also noticeable that feature-level and late fusion (averaging of classification scores) methods in two-stream networks displayed comparable validation performance in both CNN-LSTM and CNN-BiLSTM. However, the fusion of RGB video frames and dense optical flow frames at the start of the network displays degrading performance than other configurations of CNN-LSTM and CNN-BiLSTM networks. Even the two-stream CNN-LSTM with feature-level (accuracy = 95.46) and late fusion (accuracy = 95.33) methods performed relatively better than the early fusion-based CNN-BiLSTM (accuracy = 93.02). Thus, combining RGB and optical flow video frames at the start of a two-stream network has no substantial influence on the overall performance. Conclusively, the feature-level fusion of spatial and temporal streams in the CNN-BiLSTM model performed best in all experiments (f1-score = 0.9316) followed by the late fusion (f1-score = 0.9227) of two-stream networks.

Figure 7 depicts the accuracy and loss curves of the proposed two-stream deep learning framework. In general, both curves demonstrate the stability and high performance of the proposed model.

Table 6 Performance of two-stream CNN-LSTM and CNN-BiLSTM models with different fusion strategies on the YT-C2F dataset

Model		Training	Validation			
Method	Fusion strategy	Accuracy (%)	Accuracy (%)	Precision (%)	Recall (%)	F1-Score
EfficientNet-B7 + LSTM (Spatial + Motion)	Early	99.21	92.10	89.5	88.12	0.8880
	Feature	99.49	95.46	91.75	90.5	0.9112
	Late	99.30	**95.33**	**91.56**	**91.03**	**0.9129**
EfficientNet-B7 + BiLSTM (Spatial + Motion)	Early	99.30	93.02	89.97	88.32	0.8914
	Feature	99.51	**96.54**	**94.10**	**92.24**	**0.9316**
	Late	99.47	96.76	93.12	91.45	0.9227

The bold highlighted entries shows the overall best classification result against all sets of experiments

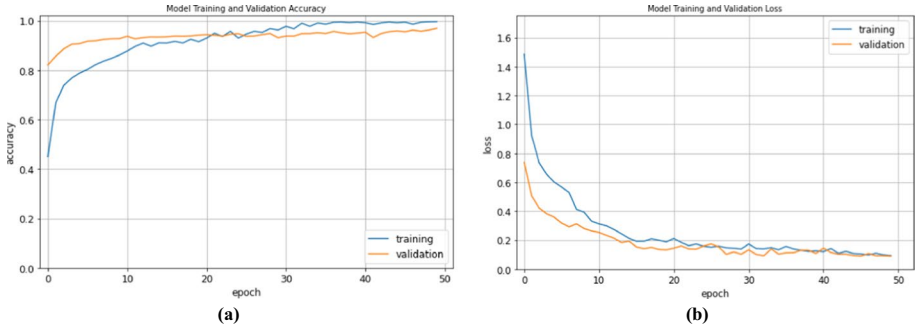


Fig. 7 Proposed model training history (**a**) Model training and validation accuracy, and (**b**) Model training and validation loss

4.4 Comparison with state-of-the-art models

This study compares the performance of the proposed framework with existing state-of-the-art models and techniques used in disturbing content detection and classification by working on different data modalities i.e., text, audio, images and videos. A comparison of the proposed work with other recent works is given in Table 7. The existing studies are categorized into single-stream and two-stream deep learning models. An interesting point to note is that the majority of the existing studies investigated video information and applied ImageNet pretrained CNN models to extract features, which are further processed by either machine learning or deep learning classifiers to perform video classification.

In contrast to existing machine learning classifiers, the proposed model outperformed single-stream AlexNet-SVM [52] by 2.44%, GoogLeNet-SVM [75] by 10.04%, and ResNet-SVM [3] by 9.1% respectively. Comparatively, the simple CNN-based neural network frameworks failed to understand the complexities of videos and reported substantially lower classification accuracy than other models [5, 37]. The most frequent technique, for disturbing video content classification, in existing studies employed an LSTM network on top of pretrained CNN features. In single-stream network configuration, the ResNet-LSTM [77] reported best accuracy ($95.5 \pm 1\%$) than VGG-LSTM [64, 70] and Inception-LSTM [57] frameworks. It even performed better than single-stream spatio-temporal VGG-BiConvLSTM [28]. This shows that the depth of neural network plays an important part in extracting the relevant video features for classification. In our previous work [85], the EfficientNet-B7 is applied in single-stream settings to allow more depth than ResNet-101 and BiLSTM is used to capture the long-term dependencies of videos by processing the visual data in both forward and backward directions. As a result, it improves accuracy over ResNet-LSTM by 0.16%. The comparison of single-stream and two-stream neural network models asserted that two-stream models perform better, and generate high accuracy scores on visual data. This indicates that motion information contains strong representations and leads to better neural network frameworks. Furthermore, the proposed two-stream EfficientNet-BiLSTM network with feature-level fusion method outperformed other existing two-stream models including CNN-LSTM [31] and CNN-SVM [32, 59] by achieving a validation accuracy of 96.54%. The performance comparison highlighted the efficacy of employing the BiLSTMs on top of CNN features in two-stream network to understand the long-term global temporal dependencies of spatial and motion information by considering

Table 7 Performance comparison of our proposed method with state-of-the-art approaches

# streams	Approach	Dataset	Pretrained CNN Model	Data modality	Classification type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score
Single-Stream Deep Learning Frameworks	SEIC detection using CNN-SVM [75]	Pornography-2k	GoogLeNet	image	Binary	86.5	–	86.2	–
	Cartoon pornography detection using CNN-SVM [3]	NPDI-Cartoon	ResNet-50 EfficientNet-B7	image, video	Binary	87.44 ± 2.78	86.655	89.128	0.8612
	Image and video pornography classification using CNN-SVM ensemble model [52]	NPDI	AlexNet GoogLeNet	image, video	Binary	94.1 ± 2	–	–	–
	Child unsafe content classification using CNN [37]	YouTube Seed Video	–	video	Binary	74			
	Video violence classification using a feed-forward deep neural network [5]	VSD2014	–	video, audio	Binary	51	–	–	–
	Child unsafe video content identification from YouTube using fine-grained CNN-LSTM autoencoder [64]	YouTube Anime Video	VGG-16	video	Binary	–	Safe = 81 Unsafe = 80	Safe = 80 Unsafe = 81	–

Table 7 (continued)

# streams	Approach	Dataset	Pretrained CNN Model	Data modality	Classification type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score
	Disturbed content detection on YouTube Kids platform using multimodal CNN-LSTM [70]	YouTube Cartoon Video	VGG-19	video, audio	Multi	92.83	92.72	91.71	–
	Adult content detection using CNN-LSTM [77]	NPDI	GoogLeNet ResNet	video	Binary	95.5 ± 1 (ResNet-101)	–	–	–
	Disturbed YouTube content detection using CNN-LSTM [57]	YouTube Video	Inception-V3	statistics, image, text	Binary	84.3	82.1	89.0	0.829
	Video violence detection using BiConvLSTM [28]	Violent Flows	VGG-13	video	Binary	92.18 ± 3.29	–	–	–
	Inappropriate YouTube video content detection and classification using CNN-BiLSTM [85]	YouTube Cartoon Video	EfficientNet-B7	video	Binary	95.66	93.17	92.22	0.9267
Two-Stream Deep Learning Framework	Bloody video recognition using multimodal CNN-LSTM [31]	Bloody video database	ResNet-50	video audio	Binary	95 (feature fusion)	–	–	–

Table 7 (continued)

# streams	Approach	Dataset	Pretrained CNN Model	Data modality	Classification type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score
	Video pornography detection using CNN-SVM [59]	Pornography-2k	GoogLeNet	video	Binary	96.4 ± 1.5 (late fusion)	–	–	–
	Disturbing cartoon content detection using CNN-SVM [32]	Elsagate	NASNet	video	Binary	92.6 (late fusion)	–	–	0.887
	Disturbing video content detection and classification using two-stream CNN-BiLSTM **(proposed)**	YT-C2F	EfficientNet-B7	video	Multi	**96.54 (feature fusion)**	**94.10**	**92.24**	**0.9316**
	Disturbing video content detection and classification using two-stream CNN-BiLSTM **(proposed)**	YT-C2F	EfficientNet-B7	video	Binary	**Safe = 96.73 Unsafe = 96.35 (feature fusion)**	**Safe = 94.78 Unsafe = 93.42**	**Safe = 94.29 Unsafe = 90.18**	**0.9316**

the past and future context. A confusion matrix is used to demonstrate the classification performance, which records the correct and incorrect predictions on each class in the form of a matrix. As shown in Fig. 8, the predicted class labels are displayed in the rows of the matrix, while the actual class labels are displayed in the columns of matrix. All the correctly classified video instances of dataset are shown in diagonal values of matrix. The statistics of the confusion matrix showed that the proposed EfficientNet-BiLSTM framework can process 6 RGB video frames with the corresponding optical flows frames per second of video by discarding the redundant information, and making it a suitable candidate for classifying the visual content in real-time video processing applications.

5 The hypothesis and limitations

The proposed two-stream deep learning framework ensures that it can effectively detect and classify the disturbing content in children-specific cartoon videos on YouTube and YouTube Kids platforms. However, this work has some limitations. Due to limited available memory and computational resources, the trained model made predictions with some inference delay, as video preprocessing is a resource-intensive task that involves frame-by-frame video analysis. To provide any video as an input to the model, the static RGB video frames and their corresponding motion optical flow video frames have to be extracted first and then converted to the fixed-size feature vector. This process causes some inference delay in generating the classification prediction result of per-second information of the video. In addition, the proposed two-stream framework is trained on a specific genre of cartoon videos, which may not be an accurate representation of all children's content on YouTube.

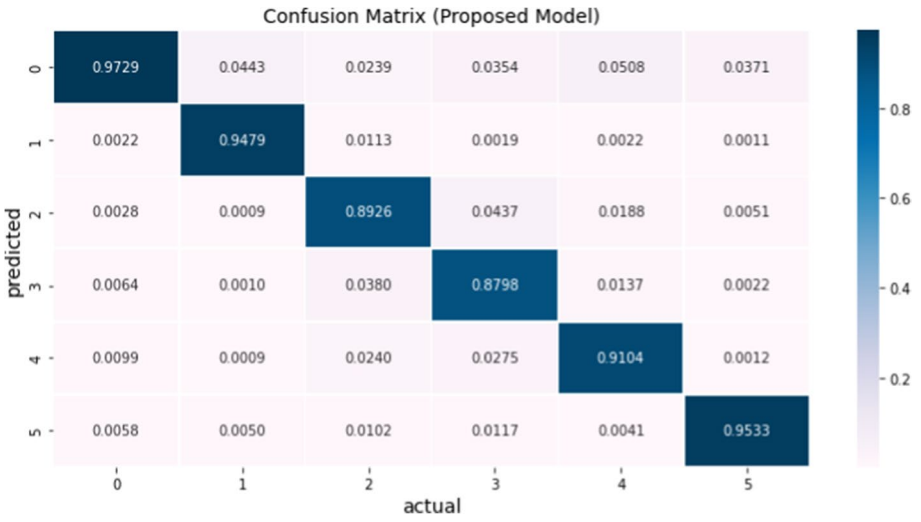


Fig. 8 Confusion matrix for the classification results obtained by the proposed two-stream EfficientNet-BiLSTM (feature-level fusion) framework

6 Conclusion and future work

In this paper, we proposed a novel end-to-end trainable two-stream EfficientNet-BiLSTM framework for disturbing video content classification. Different two-stream fusion mechanisms are applied to combine the static RGB and motion optical flow video representations. All experiments are performed on a customized large-scale YT-C2F dataset with 124 hours of YouTube cartoon videos. Results demonstrated that the proposed framework with feature-level fusion performed better (f1-score = 0.9316) than other fusion methods. Furthermore, the bidirectional LSTM has an advantage over LSTM in the proposed model for capturing the long-term temporal dependencies of spatio-temporal video descriptors. Performance comparison with various state-of-the-art models showed that adding motion information in a deep learning framework improves the classification score. To summarize, this work may help in setting parental controls on social media platforms where the disturbing visual content filters automatically through browser extensions or plugins. In future, we plan to include the other categories of YouTube videos (e.g., education, sports, health, news, movies, etc.) in the YT-C2F dataset to obtain more powerful and generic features for disturbing video content classification. The other possible direction is to apply the deep reinforcement learning method for disturbing video content classification by looking into fewer video frames rather than processing the whole video.

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Declarations

Research involving human participants and/or animals This article does not contain any studies with human participants or animals performed by any of theauthors.

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